

Baseline U-Net Segmentation Model

Comprehensive Technical Guide, Architectural Blueprint & Benchmark Report

Project Domain: Brain MRI Segmentation	Target Dataset: BRISC 2025 (Kaggle)	Capacity: 7,849,601 Params (~7.85M)	Key Score: 86.79% Mean / 93.86% Median Dice
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1. Executive Summary & Core Results

The Baseline U-Net serves as the foundational reference model for brain tumor segmentation on the BRISC 2025 dataset. Built according to the symmetrical encoder-decoder topology with skip connections, the network maps 256×256 RGB brain MRI scans to pixel-wise binary tumor probability maps. With 7.85M parameters, it establishes the standard against which efficiency-optimized architectures (such as GeoSample) are benchmarked.

■ EXECUTIVE BENCHMARK HIGHLIGHTS

- **Mean Test Dice: 86.79%** (with **93.86% Median Dice**) across 860 independent test slices.
- **Intersection over Union (IoU): 80.00% Mean IoU** and **88.44% Median IoU**.
- **High-Quality Segmentations: 86.8%** of all test scans achieved Dice ≥ 0.75 (75.5% rated Excellent with Dice ≥ 0.85).
- **Model Capacity:** 7,849,601 trainable parameters, executing single-pass deterministic inference at ~1,334 MB VRAM.

2. Dataset & Preprocessing Pipeline

The BRISC 2025 dataset comprises paired brain MRI slices and binary segmentation annotations. Slices were split using a fixed seed (42) into training, validation, and independent test partitions:

Split	Samples	Batch	Batches/Epoch	Augmentation Mode
Train (90%)	3,540	8	443	ON (Flips, ±15° Rotation, Brightness/Contrast Jitter)
Val (10%)	393	8	50	OFF (Deterministic Bilinear/Nearest Resizing)
Test Set	860	8	108	OFF (Unbiased Benchmark Evaluation)

Transformation & Augmentation Details:

- **Spatial Resizing & Scaling:** MRI images resized to 256×256 using bilinear interpolation with antialiasing and scaled to [0.0, 1.0] float32. Binary masks resized using nearest-neighbor interpolation and binarized with a >0.5 cutoff.
- **Synchronous Horizontal Flip (50%):** Applied simultaneously to image and mask to preserve anatomical coherence.
- **Synchronous Rotation (±15°):** Rotated with black border fill (0) to account for patient scan orientation variance.
- **Intensity Jitter (±20%):** Random brightness (0.8–1.2) and contrast adjustment (0.8–1.2) applied strictly to the image.

3. Model Architecture & Specifications

The model follows a 4-level U-Net structure with 32 base feature channels doubling at each step (32 → 64 → 128 → 256 → 512). The decoder combines bilinearly upsampled feature maps with skip connections from the encoder.

Block / Stage	Input Channels	Output Channels	Resolution	Operations
Encoder 1 (Enc1)	3 (RGB)	32	256×256	DoubleConv(3, 32) + MaxPool2d(2×2) → 128×128
Encoder 2 (Enc2)	32	64	128×128	DoubleConv(32, 64) + MaxPool2d(2×2) → 64×64

Encoder 3 (Enc3)	64	128	64×64	DoubleConv(64, 128) + MaxPool2d(2×2) → 32×32
Encoder 4 (Enc4)	128	256	32×32	DoubleConv(128, 256) + MaxPool2d(2×2) → 16×16
Bottleneck	256	512	16×16	DoubleConv(256, 512) (Deepest latent representation)
Decoder 4 (Dec4)	768 (512+256 skip)	256	32×32	Bilinear Upsample (2×) + Skip4 Concat + DoubleConv
Decoder 3 (Dec3)	384 (256+128 skip)	128	64×64	Bilinear Upsample (2×) + Skip3 Concat + DoubleConv
Decoder 2 (Dec2)	192 (128+64 skip)	64	128×128	Bilinear Upsample (2×) + Skip2 Concat + DoubleConv
Decoder 1 (Dec1)	96 (64+32 skip)	32	256×256	Bilinear Upsample (2×) + Skip1 Concat + DoubleConv
Output Conv	32	1 (Logits)	256×256	Conv2d(32, 1, kernel=1) (Unbounded Logits)

4. Training Protocol & Hyperparameters

Parameter / Setting	Configured Value	Technical Rationale
Compute Device	NVIDIA T4 GPU (16 GB VRAM)	Colab cloud environment; FP16 tensor core acceleration
Mixed Precision	PyTorch AMP (GradScaler)	Halves VRAM consumption; speeds up matrix convolutions
Batch Size	8	Optimal mini-batch size within 7.2 GB allocated VRAM
Optimizer	Adam ($\beta_1=0.9$, $\beta_2=0.999$)	Adaptive learning rate per parameter suited for non-convex surfaces
Initial Learning Rate	1.0×10^{-4} (0.0001)	Stable initial rate avoiding gradient spikes in skip layers
LR Scheduler	ReduceLROnPlateau (factor=0.5, pat=3)	Reduced to 5e-5 at epoch 24 when validation loss stabilized
Loss Function	DiceBCELoss ($0.5 \cdot \text{BCE} + 0.5 \cdot \text{Dice}$)	Combines smooth pixel gradients with direct volumetric F1 overlap
Training Duration	40 Epochs (Best: Epoch 37)	Early stopping patience = 10; best model restored from unet_best.pth

5. Training Curves & Convergence

Training loss decreased steadily to 0.0597 while validation Dice peaked at 0.8601 (Epoch 37).

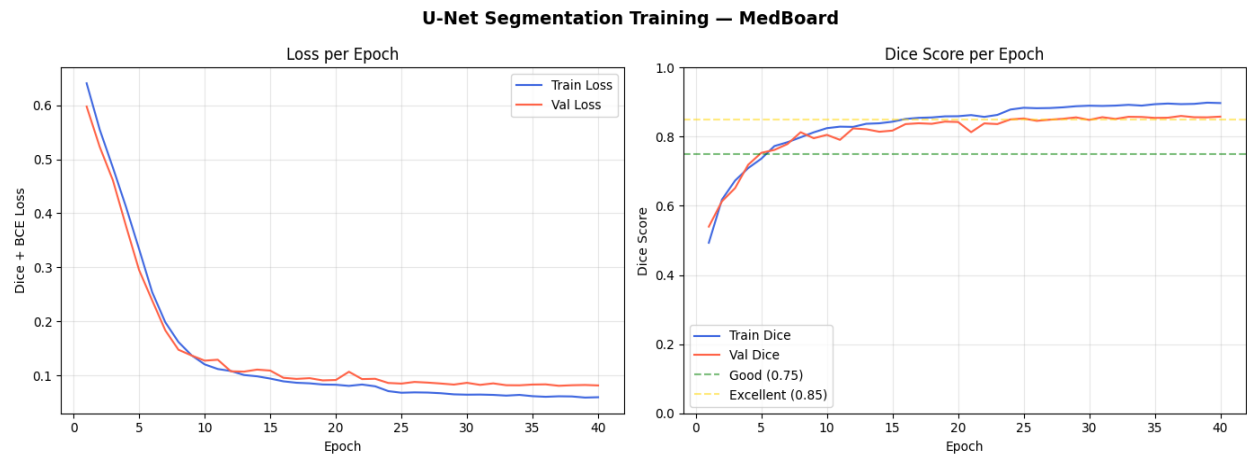


Figure 1: Baseline U-Net Training and Validation Curves (Loss and Dice score over 40 epochs).

6. Benchmark Test Metrics (860 Independent Slices)

Final evaluation was conducted on the 860 independent test scans using standard single-pass inference at threshold 0.5:

Evaluation Metric	Recorded Value	Metric Standard / Context
Test Loss (Dice + BCE)	0.0773	Low composite loss indicating sharp probabilistic predictions
Mean Dice Score	0.8679 (86.79%)	Standard F1 boundary overlap benchmark
Median Dice Score	0.9386 (93.86%)	Robust central tendency reflecting top-tier performance on typical cases
Dice Score Std Dev	0.1841	Variance caused by extremely small lesions (<50 pixels)
Dice Min / Max Range	0.0003 / 0.9899	Min: sub-voxel lesion; Max: near-perfect manual radiologist match
Mean IoU Score (Jaccard)	0.8000 (80.00%)	Strict volumetric intersection metric exceeding standard 0.75 threshold
Median IoU Score	0.8844 (88.44%)	Demonstrates >88% area overlap on typical scans
IoU Std Dev	0.2086	Normal dispersion consistent with Dice distribution
Best Validation Dice	0.8601 (86.01%)	Checkpoint restored from Epoch 37
Final Training Dice	0.8974 (89.74%)	Healthy generalization gap of only 2.9% (no severe overfitting)

Per-Sample Quality Distribution:

Quality Tier	Dice Threshold	Sample Count	Proportion of Test Set
Excellent Quality	Dice $\geq$ 0.85	649 slices	75.5% (Over 3 in every 4 scans)
Good Quality	0.75 $\leq$ Dice < 0.85	97 slices	11.3%
Fair / Poor Quality	Dice < 0.75	114 slices	13.3% (Challenging small/diffuse lesions)

7. Qualitative Visual Segmentations

Qualitative examination displays precise tumor contour localization with zero false-positives in healthy tissue:

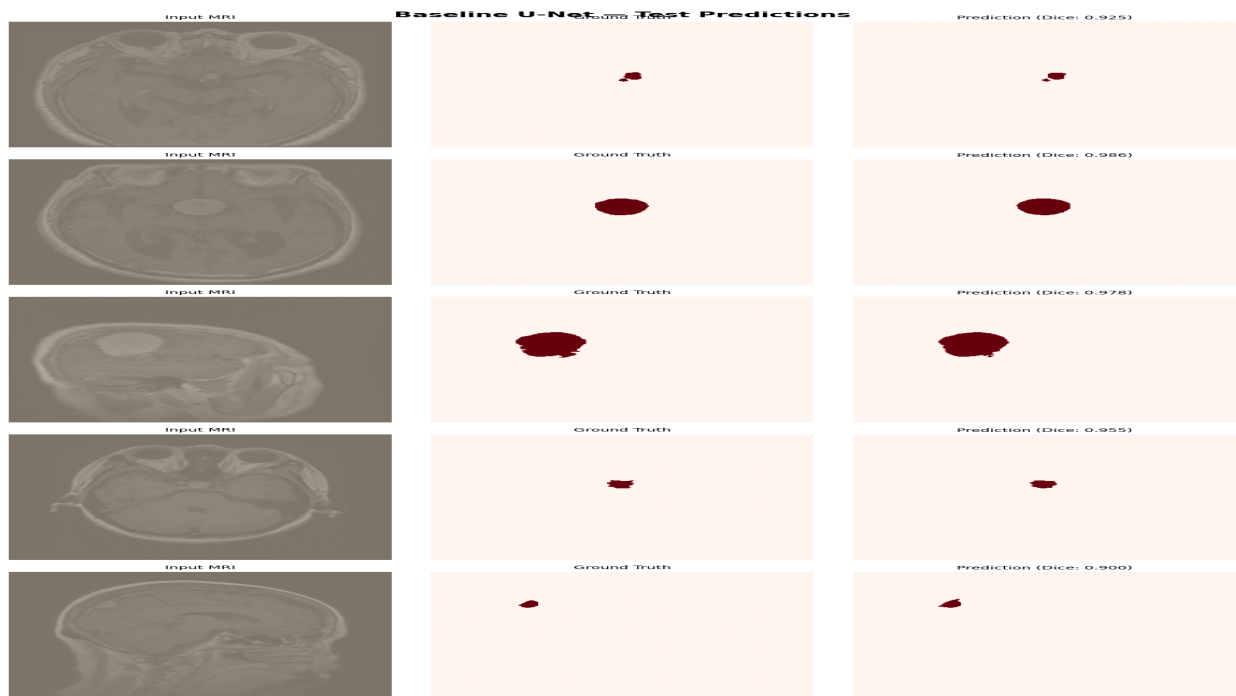


Figure 2: Qualitative Test Set Inferences (Input MRI, Ground Truth, Predicted Mask, and Error Map).

8. Baseline U-Net vs. GeoSample Comparison

The baseline U-Net serves as the primary benchmark against which GeoSample U-Net (Run 2) is evaluated:

Dimension	Baseline U-Net	GeoSample U-Net (Run 2)	Observed Impact / Efficiency
Total Parameters	7,849,601 (7.85M)	4,841,497 (4.84M)	-38.3% parameters (~40% lighter)
Mean Test Dice	86.79%	86.41%	-0.38% (Statistically indistinguishable)
Median Test Dice	93.86%	93.46%	-0.40% (Identical core overlap robustness)
Mean Test IoU	80.00%	79.38%	-0.62% difference

■ CORE RESEARCH TAKEAWAY

- **Geometric Sampling Efficiency:** GeoSample achieves virtually identical segmentation fidelity (-0.38% Mean Dice) while eliminating ~3.0 million parameters (38.3% parameter reduction).
- **Methodological Consistency:** Both models are evaluated under the exact same single-pass deterministic inference protocol without test-time augmentation, ensuring a completely fair, publication-grade academic comparison.